

5.3.2 The hormonal control of blood glucose and the management of diabetes

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) (i) the structure and function of the pancreas as an endocrine gland	To include the use of diagrams, photomicrographs and electron micrographs to identify the Islets of Langerhans and non-endocrine tissues in the pancreas.
(ii) practical observations of prepared slides of pancreatic tissue using a light microscope	<i>M0.1, M0.2, M1.8</i> PAG1 HSW4
(b) the regulation of blood glucose by negative feedback	To include details of the action of insulin and glucagon at a cellular level.
(c) the different types of diabetes	To include the causes, risk factors and diagnosis of Type 1 and Type 2 diabetes.
(d) the fasting blood glucose test, glucose tolerance testing and the use of biosensors in the monitoring of blood glucose concentrations	To include the analysis and evaluation of secondary data (from glucose tests). <i>M0.1, M0.2, M1.7</i> HSW3, HSW4, HSW5, HSW6, HSW9, HSW10, HSW11
(e) the treatment and management of Type 1 and Type 2 diabetes	To include the use of insulin, lifestyle interventions, the use of drugs and the monitoring of glucose control using glycosylated haemoglobin concentrations. <i>M0.1, M0.2, M0.3, M1.1, M1.5</i>
(f) the team of health professionals involved in the management of diabetes (i.e. the diabetes nurse, dieticians, retinal screeners, podiatrists), including the role of evidence based practice	
(g) the future impact of diabetes on the human population.	